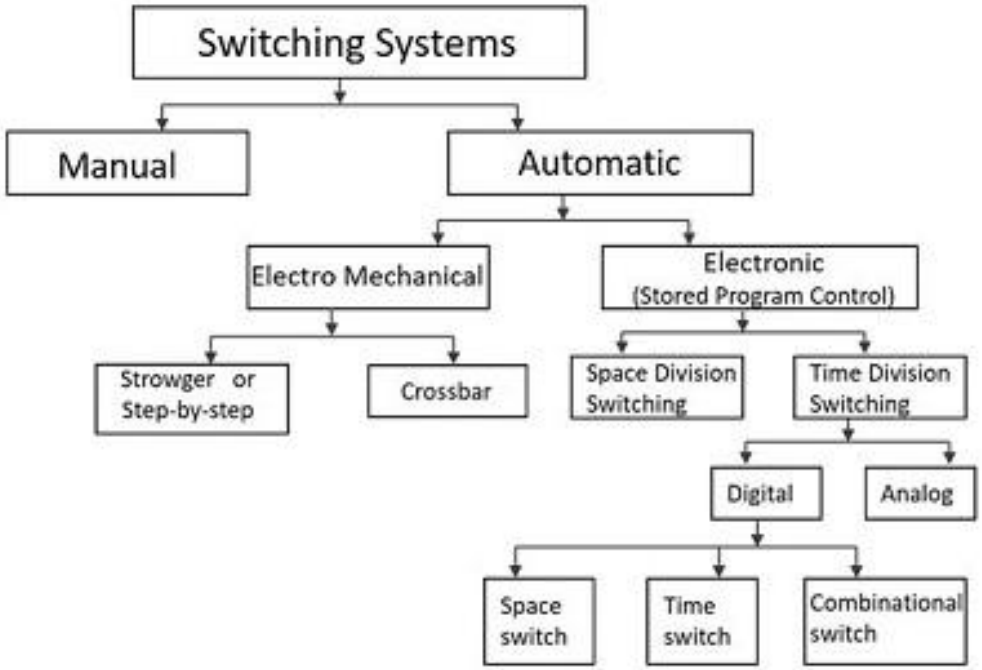
Telecommunication Systems
ICE 3109

Lecture 5
Switching in Telecommunication

Switching Systems

- A Switching system can be understood as a collection of switching elements arranged and controlled in such a way as to set up a common path between any two distant points.
- The introduction of switching systems reduced the complexity of wiring and made the telephony hassle-free.

Switching Systems



Step-by-Step Switching

- Step-by-step switching is a fundamental concept in the history of telecommunications, particularly in the development of telephone systems.
- It refers to the method by which telephone calls are connected through a series of switches that work sequentially, ensuring that the call reaches its intended destination.
- This method was widely used before the advent of more advanced digital switching systems.

Step-by-Step Switching



Picture of a Western Electric Step-by-Step Community Dial Office switch at the Museum of Communications (Seattle, WA)

Background

- **Early Telephone Systems:** Initially, telephone systems were manually operated, with operators connecting calls by plugging wires into switchboards.
- As demand grew, the need for automation led to the development of electromechanical switching systems

Switchboard operator

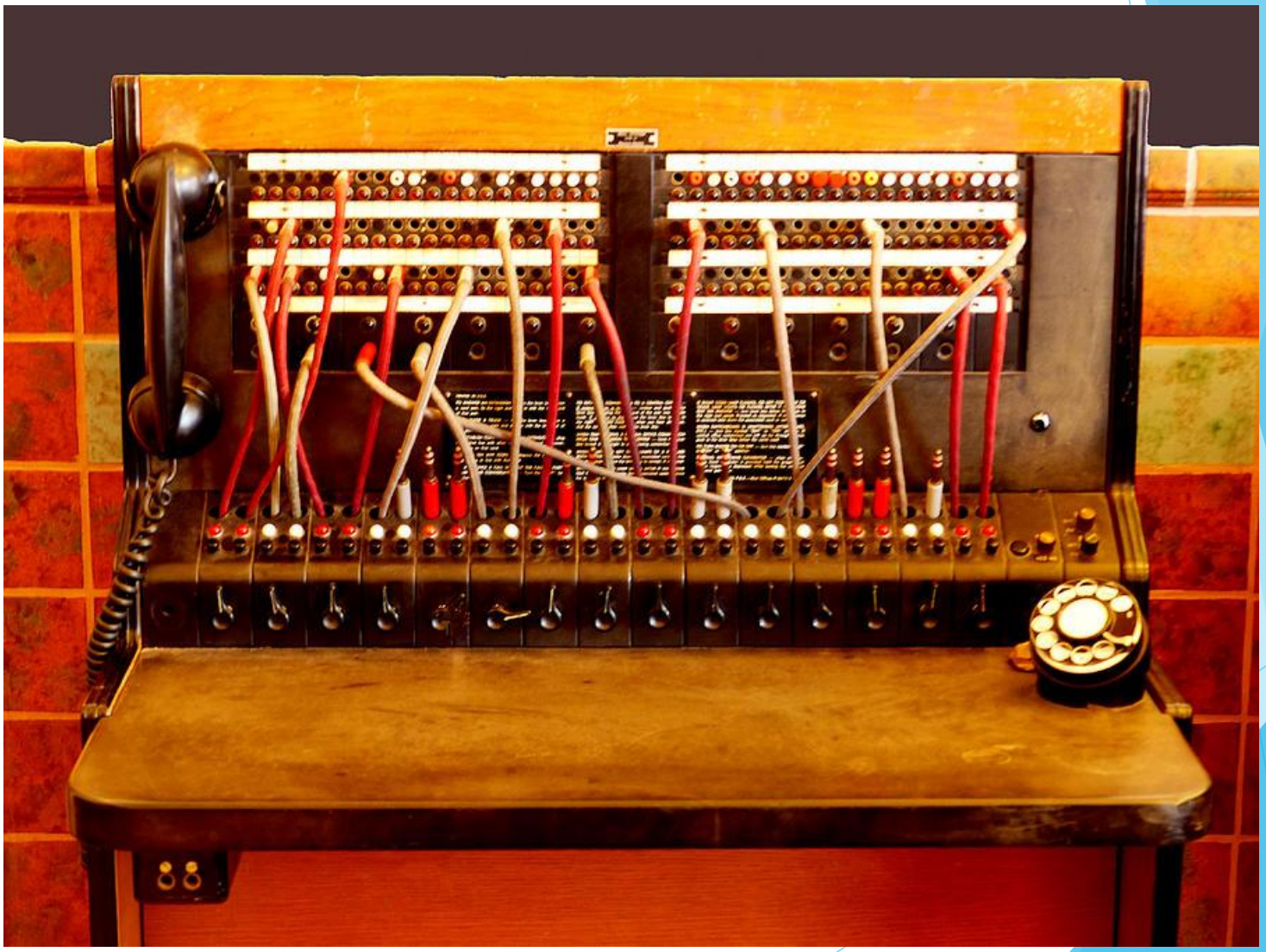
- In the early days of telephony, companies used manual telephone switchboards, and switchboard operators connected calls by inserting a pair of phone plugs into the appropriate jacks.
- They were gradually phased out and replaced by automated systems, first those allowing direct dialing within a local area, then for long-distance and international direct dialing.

Switchboard operator



Seattle telephone operators in a private branch exchange in 1952

Telephone Switchboard



Telephone Switchboard

- A typical manual telephone switchboard has a vertical panel containing an array of jacks with a desk in front.
- The desk has a row of switches and two rows of plugs attached to cables that retract into the desk when not in use.
- Each pair of plugs was part of a cord circuit with a switch associated that let the operator participate in the call or ring the circuit for an incoming call.
- Each jack had a light above it that lit when the customer's telephone receiver was lifted (the earliest systems required the customer to hand-crank a magneto to alert the central office and, later, to "ring off" the completed call).

Switchboard operator

- Lines from the central office were usually arranged along the bottom row.
- Before the advent of operator distance dialing and customer direct dial (DDD) calling, a switchboard operator would work with their counterparts in distant central office to complete long-distance calls.
- Switchboard operators are typically required to have very strong communication skills.

Switchboard operator

- Before the advent of automatic exchanges, an operator's assistance was required for anything other than calling telephones across a shared party line.
- Callers spoke to an operator at a central office who then connected a cord to the proper circuit in order to complete the call.
- Being in complete control of the call, the operator was in a position to listen to private conversations

Telephone Switchboard

- Automatic or dial systems were developed in the 1920s to reduce labor costs as usage increased, and to ensure privacy to the customer.
- As phone systems became more sophisticated, less direct intervention by the telephone operator was necessary to complete calls.
- With the development of computerized telephone dialing systems, many telephone calls which previously required live operators could be placed directly by calling parties without additional human intervention.

Telephone Switchboard

- As well as the people that were employed by the public networks, operators were required at private branch exchanges (PBX) to answer incoming calls and connect them to the correct extensions.
- Today, most large organizations have *direct inward dialing*, or *direct dial-in*.
- Smaller workplaces may have an automated system which allows callers to enter the extension number of the called party, or a receptionist who answers calls and performs operator duties.

Telephone Switchboard

- Depending on the employment setting, the roles and level of responsibilities of a PBX operator can vary greatly, from performing wake-up calls in a hotel to coordinating emergency responses, dispatching, and overhead paging in hospitals.
- Operators employed in healthcare settings may have other duties, such as data entry, greeting patients and visitors, taking messages, triaging, or performing after-hours answering service. Experienced, well-trained operators generally command higher salaries.

Telephone Switchboard

- In January 1878 George Willard Croy became the world's first telephone operator when he started working for the Boston Telephone Dispatch company.
- Emma Nutt became the first female telephone operator on 1 September 1878 when she started working for the Boston Telephone Dispatch company.
- Emma was hired by Alexander Graham Bell and, reportedly, could remember every number in the telephone directory of the New England Telephone Company.

Step-by-Step Switching

- A step-by-step switching is also called the Direct control system as the relevant signaling tones are sent out to the subscriber by the switching elements or selectors at the appropriate stages of switching.
- Almon B. Strowger, an American inventor, invented the step-by-step switch in the late 19th century. He was motivated by a desire to avoid operator interference, which he believed was costing him in business.

Step-by-Step Switching



Step-by-Step Switching Operation

- **Sequential Operation:** The core idea of step-by-step switching is that each dialed digit of a phone number directly controls a specific switch. Each switch, in turn, is responsible for routing the call to the next stage in the network.

- **Switching Stages:** Typically, a step-by-step system has multiple stages (e.g., selector switches). Each stage is responsible for a part of the phone number, and the call progresses through these stages until it reaches the final connection point.

Key Components

- **Rotary Dial:** When a caller dials a number, each pulse from the rotary dial advances a selector in the exchange.
- **Selectors:**
  - **Line Selector:** The first selector that responds to the initial digits dialed and connects to a specific line.
  - **Group Selector:** Subsequent selectors that refine the connection based on additional digits dialed.
  - **Final Selector:** The last selector that connects to the called party's line.
- **Switching Matrix:** A grid-like matrix where each intersection can connect a call by advancing selectors step-by-step until the connection is complete.

Step-by-Step Process

- **Dialing the Number:** The user picks up the phone and dials a number.
- **Advancing Selectors:** Each digit dialed advances the selectors in the exchange through its steps.
- **Establishing Connection:** The selectors navigate through the switching matrix, connecting various stages until the circuit to the called party's line is complete.
- **Call Completion:** Once the circuit is established, the two parties are connected, and they can communicate.

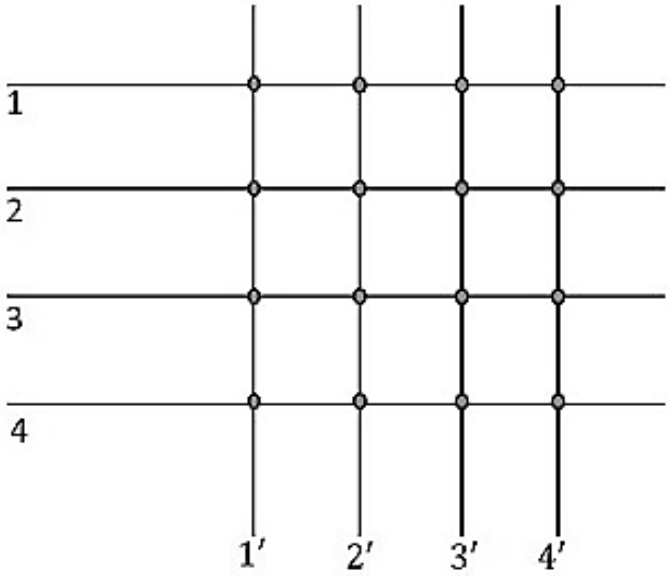
Advantages and Disadvantages

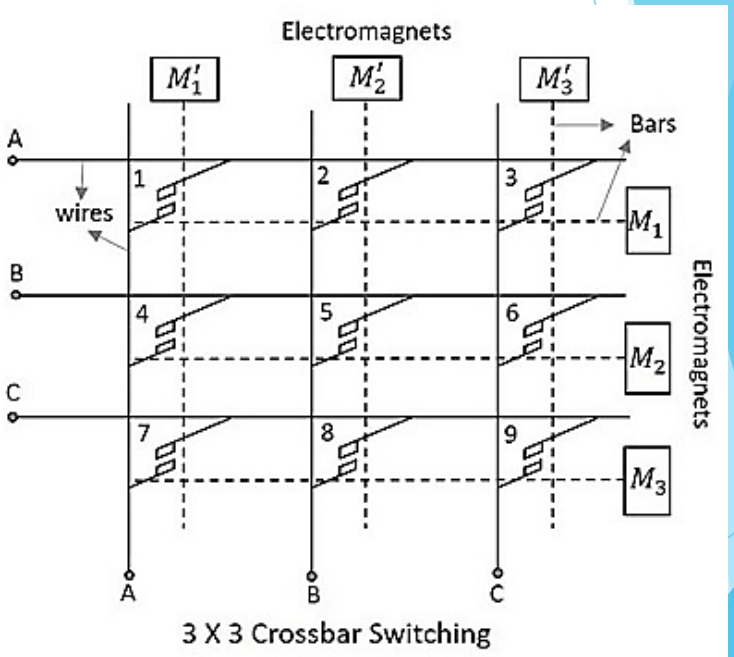
- **Advantages:**
  - Eliminated the need for manual operators for routine call connections.
  - Allowed for direct dialing by users, speeding up the process.
- **Disadvantages:**
  - Complex and prone to mechanical wear and tear due to the moving parts.
  - Scalability issues as the demand for telephone lines increased.
  - Eventually replaced by more advanced technologies like crossbar switching and digital exchanges.

Crossbar Switching

- Crossbar switching is a type of switching system used in telecommunications that utilizes a grid of horizontal and vertical bars (called crosspoints) to establish a connection between multiple inputs and outputs.
- This technology was widely used in telephone exchanges before the advent of digital switching systems.

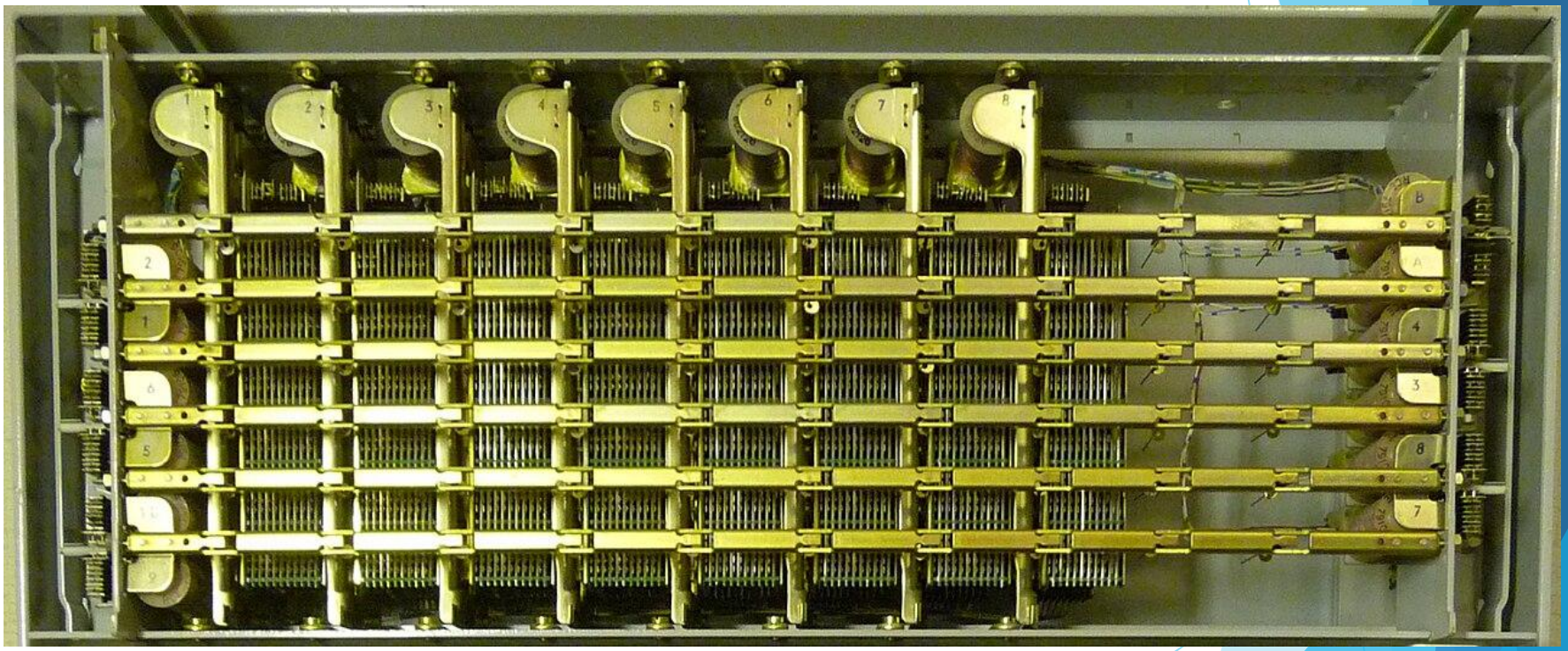
Crossbar Switching





3 X 3 Crossbar Switching

Crossbar Switching



Key Features of Crossbar Switching:

- **Grid Structure:**
  1. The crossbar switch consists of a matrix (grid) with horizontal and vertical bars. Each horizontal bar is connected to an input line, while each vertical bar is connected to an output line.
  2. At each intersection of the grid, there is a crosspoint switch that can be opened or closed to create a connection between the corresponding input and output.
- **Crosspoint Activation:**
  1. A connection is established when a crosspoint switch at the intersection of a specific horizontal and vertical bar is activated (closed). This creates a path for the signal to travel from the input line to the output line.

Key Features of Crossbar Switching:

▶ **Non-blocking:**

•The crossbar switch is typically non-blocking, meaning that any input can be connected to any output without interfering with other connections, as long as there is an available crosspoint.

▶ **Control:**

•The activation of the crosspoints is controlled by a controller, which manages the routing of calls. When a call is initiated, the controller identifies an available path and activates the corresponding crosspoints.

Key Features of Crossbar Switching:

- **Scalability:**
- The scalability of crossbar switches is limited by the number of crosspoints needed. For N inputs and M outputs, the switch requires N×M crosspoints. This makes the design complex and expensive as the number of inputs and outputs increases.

- **Historical Use:**
- Crossbar switches were widely used in early telephone exchanges due to their ability to handle multiple connections simultaneously. They were an improvement over earlier mechanical switching systems.

Advantages

- **Simplicity:** Crossbar switches are relatively simple in their operation and design.
- **Speed:** They offer fast switching times since connections are established directly at the crosspoints.
- **Reliability:** With fewer moving parts compared to older mechanical switches, crossbar systems were more reliable.

Disadvantages:

- **Scalability:** The number of crosspoints increases rapidly as the number of inputs and outputs grows, leading to increased cost and complexity.

- **Space and Power Consumption:** As the size of the matrix increases, so do the physical space and power requirements.

Transition to Digital Switching

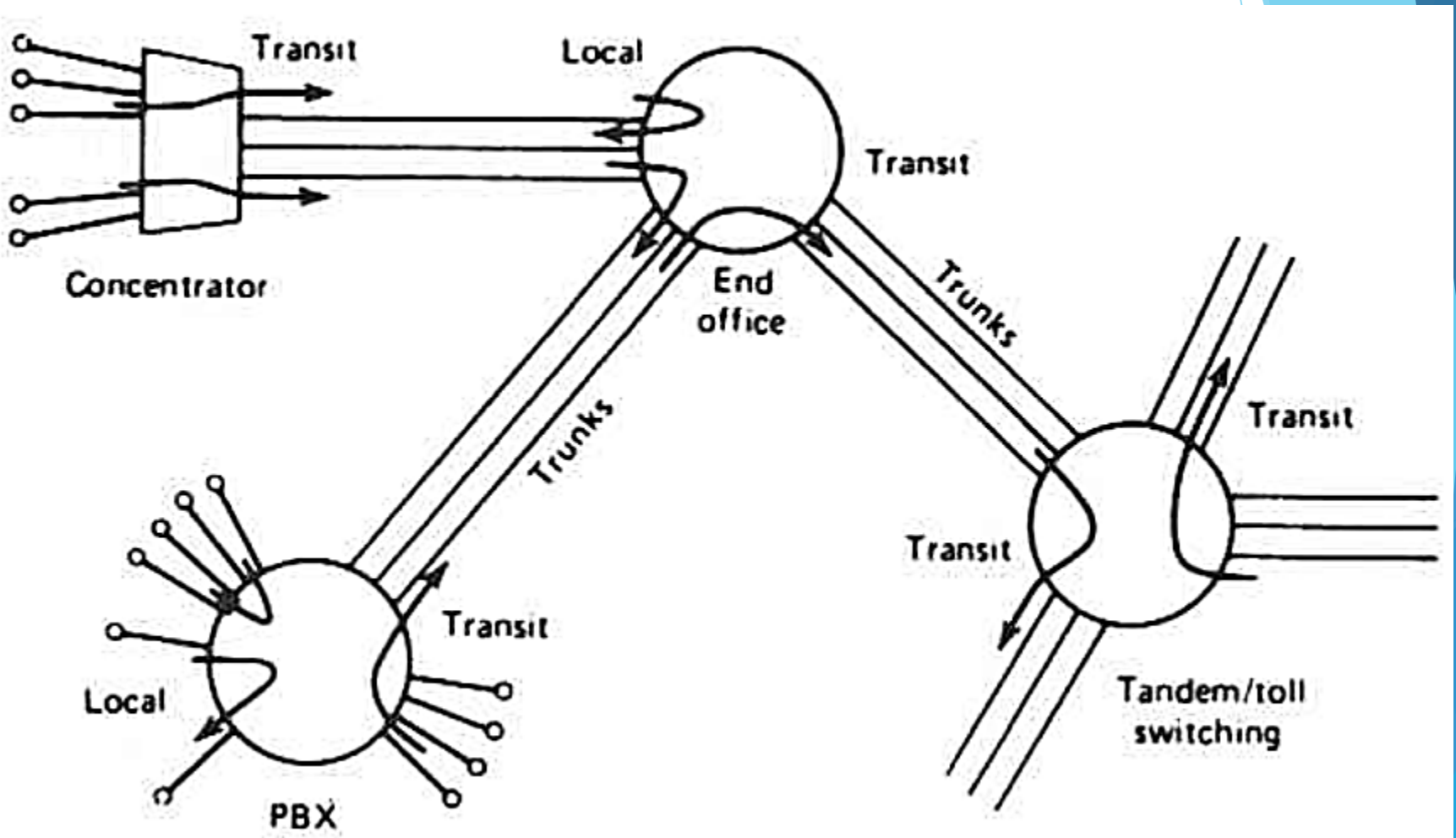
- As telecommunications evolved, crossbar switching systems were gradually replaced by digital switching systems, which offer greater flexibility, efficiency, and scalability.
- Digital systems use time-division multiplexing (TDM) and packet switching to route calls, making them more suitable for the modern telecommunications environment.
- In summary, crossbar switching was a crucial step in the evolution of telecommunication switching systems, providing a reliable and efficient method for connecting multiple lines in telephone exchanges.

Switching Functions

Obviously, the basic function of any switch is to set up and release connections between transmission channels on an "as-needed basis." The structure and operation of a switch depend on particular applications. Three switching categories for voice circuits are local (line-to-line) switching, transit (tandem) switching, and call distribution.

The most common switching function involves direct connections between subscriber loops at an end office or between station loops at a PBX. These connections inherently require setting up a path through the switch from the originating loop to a specific terminating loop. Each loop must be accessible to every other loop. This level of switching is sometimes referred to as line switching.

Switching Functions



Space Division Switching

- **Space Division Switching** is a type of switching technique used in telecommunications, where physical paths (or spaces) are used to separate multiple communication channels. It is commonly employed in traditional circuit-switched networks, such as telephone systems, to create direct physical connections between input and output lines for data transmission.

Key Concepts of Space Division Switching:

- Dedicated Physical Paths
- Crossbar Switches
- Non-blocking vs. Blocking Switching

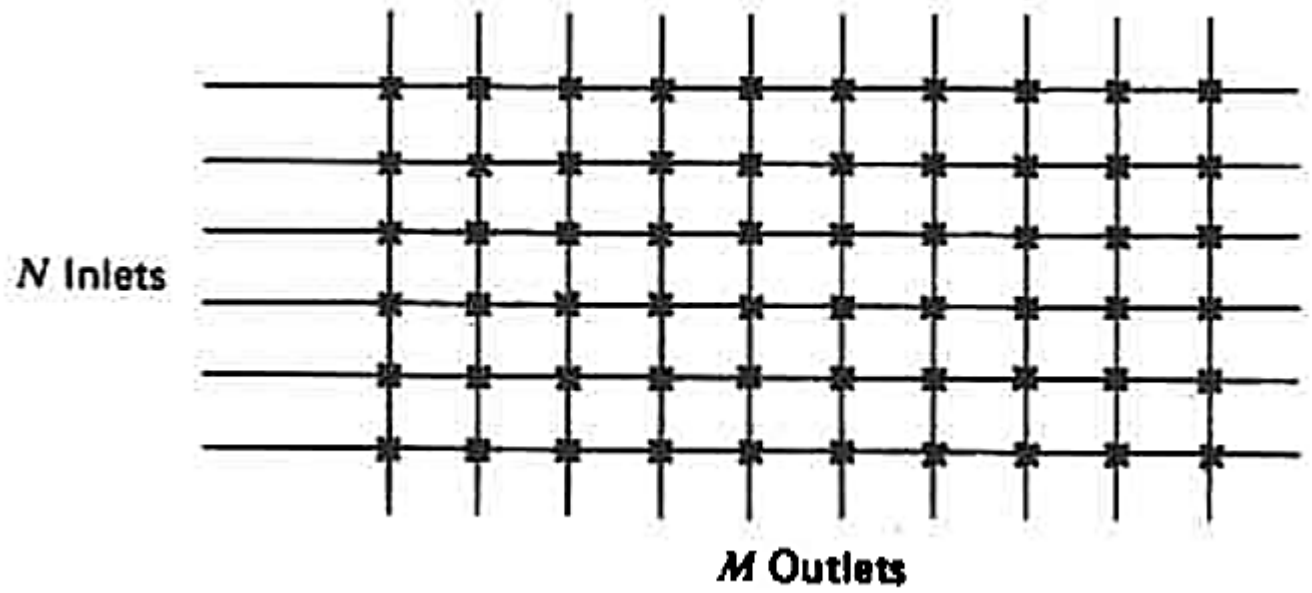
Dedicated Physical Path

- In space division switching, separate physical pathways are established for each communication channel.
- This means that every connection between an input and an output occupies a specific route, and no other communication uses that route at the same time.
- This is different from time-based or frequency-based switching, where multiple signals share the same physical path using different times or frequencies.

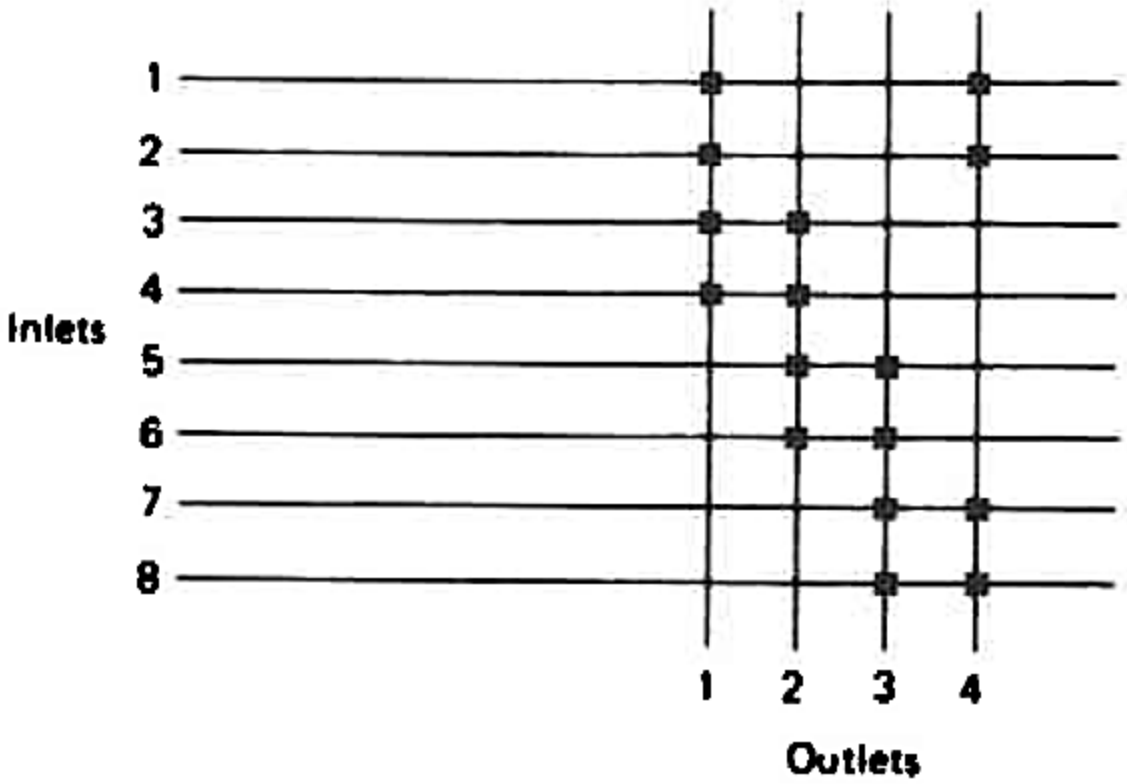
Crossbar Switches

- One of the most common implementations of space division switching uses crossbar switches. These switches are grids of horizontal and vertical wires with a switch at each intersection.
- When a connection is needed between an input and an output, the corresponding switch at the intersection is closed, creating a direct physical link between the two.
- This structure supports multiple connections as long as they don’t interfere with each other on the same physical line.

Crossbar Switches



Rectangular Crosspoint Array



Graded Rectangular Crosspoint Array

Non-blocking vs. Blocking Switching

- Non-blocking: In a non-blocking space division switch, any input can be connected to any output without affecting other ongoing connections. This is ideal for high-performance systems but is costly in terms of the number of switches needed.
- Blocking: In a blocking space division switch, some connections might not be established if certain paths are already in use by other connections, meaning resources are shared in a way that can cause contention.

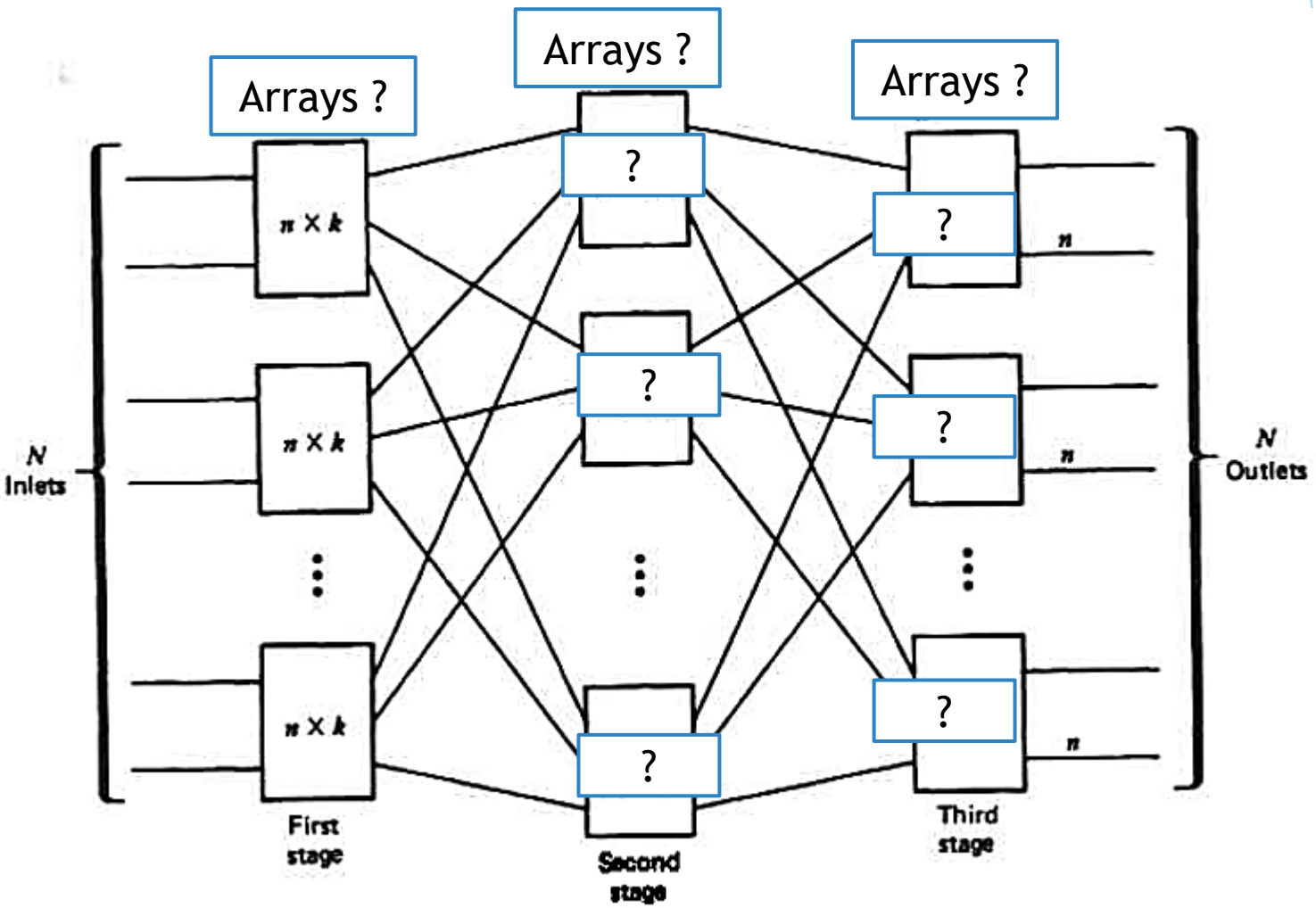
Multiple Stage Switching

In the switching structures described to this point, an inlet is connected directly to an outlet through a single crosspoint. (Four-wire switches use two crosspoints per connection, but only one for an inlet-to-outlet connection.) For this reason, these switching structures are referred to as "single-stage" switches. Single-stage switches have the property that each individual crosspoint can only be used to interconnect one particular inlet–outlet pair. Since the number of inlet–outlet pairs is equal to $N(N-1)/2$ for a triangular array, and $N(N-1)$ for a square array, the number of crosspoints required for a large switch is prohibitive. Furthermore, the large number of crosspoints on each inlet and outlet line imply a large amount of capacitive loading on the message paths. Another fundamental deficiency of single-stage switches is that one specific crosspoint is needed for each specific connection. If that crosspoint fails, the associated connection cannot be established. (An exception is the square, two-wire switch

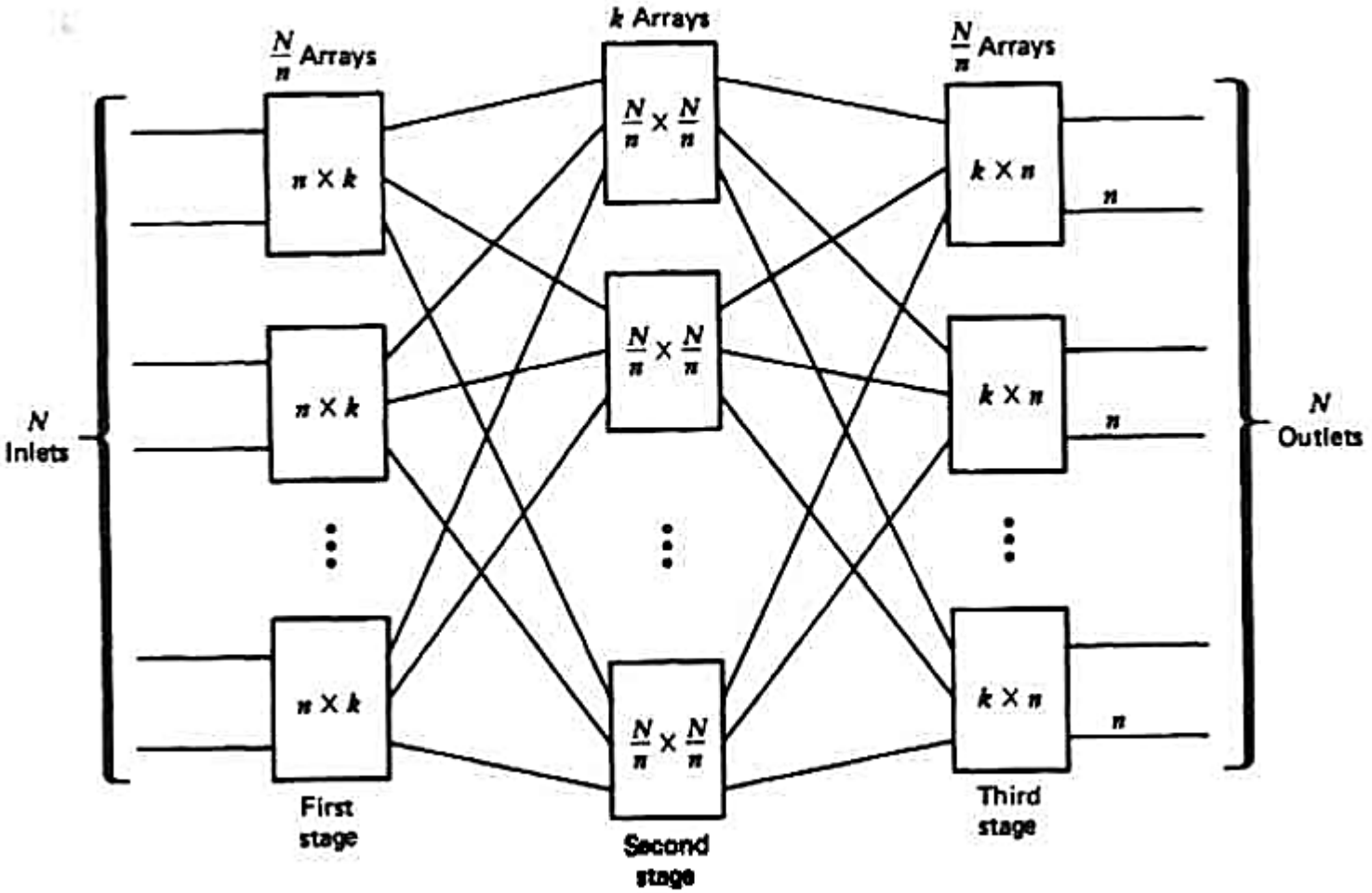
Multiple Stage Switching

Analysis of a large single-stage switch reveals that the crosspoints are very inefficiently utilized. Only one crosspoint in each row or column of a square switch is ever in use, even if all lines are active. To increase the utilization efficiency of the crosspoints, and thereby reduce the total number, it is necessary that any particular crosspoint be usable for more than one potential connection. If crosspoints are to be shared, however, it is also necessary that more than one path be available for any potential connection so that blocking does not occur. The alternate paths serve to eliminate or reduce blocking and also to provide protection against failures. The sharing of crosspoints for potential paths through the switch is accomplished by multiple-stage switching. A block diagram of one particular form of a multiple-stage switch is shown in Figure 5.6.

Three-stage Switching



Three-stage Switching



Three-stage Switching

- Number of crosspoints N_x required by a three-stage switch can be calculated as:

- $N_x = 2Nk + k\left(\frac{N}{n}\right)^2$

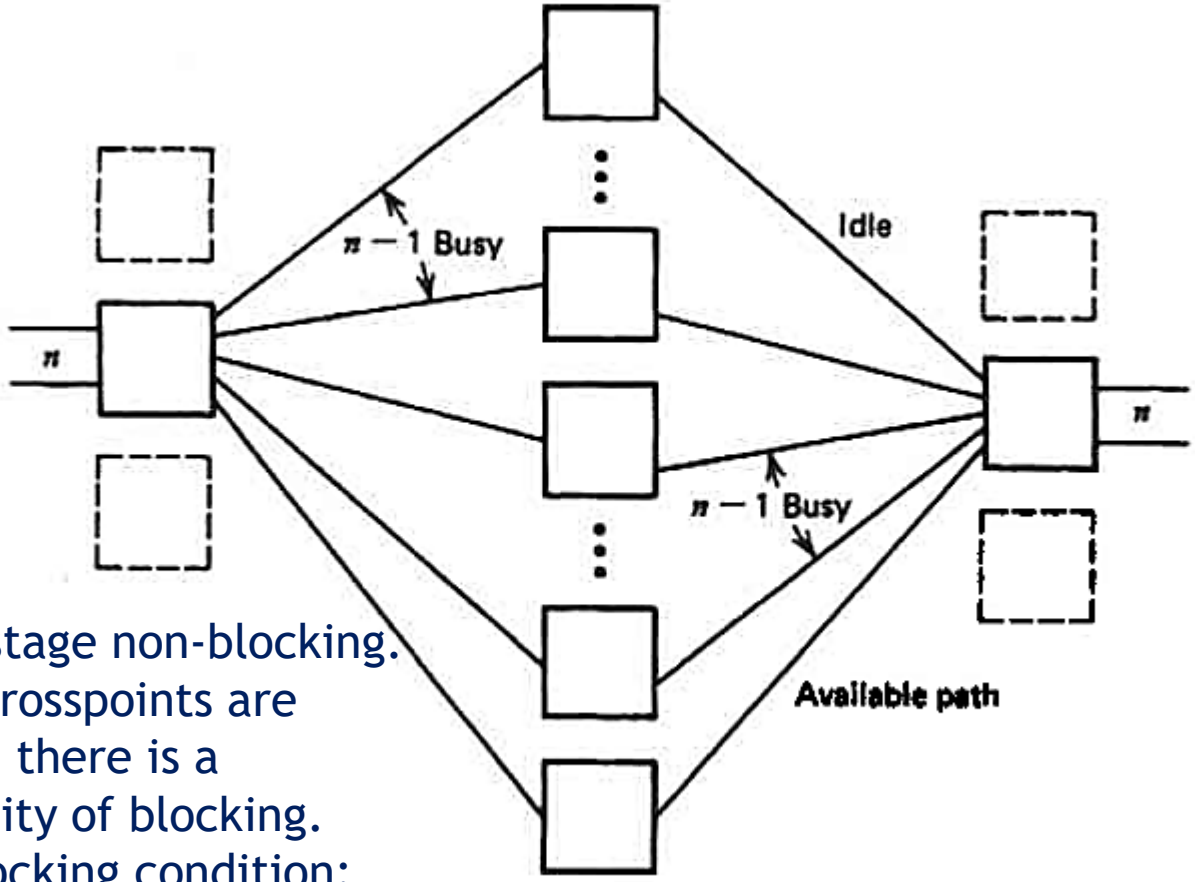
- Where

  $N =$ no. of inlets and outlets

  $n =$ size of each inlet / outlet

  $k =$ no. of center-stage arrays

Non-Blocking Three-stage Switching



- Single stage non-blocking.
- When crosspoints are shared, there is a possibility of blocking.
- Non-blocking condition:

$$k = n - 1$$

Non-Blocking Three-stage Switching

- Number of crosspoints N_x required by a three-stage switch can be calculated as:

- $N_x = 2N(2n-1) + (2n-1)\left(\frac{N}{n}\right)^2$

- Where

  $N =$ no. of inlets and outlets

  $n =$ size of each inlet / outlet

Non-Blocking Switching

TABLE 5.1 Crosspoint Requirements of Nonblocking Switches

Number of Lines	Number of Crosspoints for Three-Stage Switch	Number of Crosspoints for Single-Stage Switch
128	7,680	16,256
512	63,488	261,632
2,048	516,096	4.2 million
8,192	4.2 million	67 million
32,768	33 million	1 billion
131,072	268 million	17 billion

Blocking Probabilities

Strictly nonblocking switches are rarely needed in most voice telephone networks. Both the switching systems and the number of circuits in interoffice trunk groups are sized to service most requests as they occur, but economics dictates that network implementations have limited capacities that are occasionally exceeded during peak-traffic situations. Equipment for the public telephone network is designed to provide a certain maximum probability of blocking for the busiest hour of the day. The value of this blocking probability is one aspect of the telephone company's grade of service. (Other aspects of grades of service are availability, transmission quality, and delay in setting up a call.)

Blocking Probabilities

A typical residential telephone is busy 5–10% of the time during the busy hour. Business telephones are often busy for a larger percentage of their busy hour (which may not coincide with a residential busy hour). In either case, network-blocking occurrences on the order of 1%* during the busy hour do not represent a significant reduction in the ability to communicate since the called party is much more likely to have been busy anyway. Under these circumstances, end office switches and, to a lesser degree, PBXs can be designed with significant reductions in crosspoints by allowing acceptable blocking probabilities.

Blocking Probabilities

There are a variety of techniques that can be used to evaluate the blocking probability of a switching matrix. These techniques vary according to complexity, accuracy, and applicability to different network structures. One of the most versatile and conceptually straightforward approaches of calculating blocking probabilities involves the use of probability graphs as proposed by C. Y. Lee [3]. Although this technique requires several simplifying approximations, it can provide reasonably accurate results, particularly when comparisons of alternate structures are more important than absolute numbers. The greatest value of this approach lies in the ease of formulation and the fact that the formulas directly relate to the underlying network structures. Thus the formulations help provide insight into the network structures and how these structures might be modified to change the performance.

Blocking Probabilities : Lee Graphs

In the following analyses we are determining the blocking probabilities of various switching structures using utilization percentages, or "loadings," of individual links. The notation p will be used, in general, to represent the fraction of time that a particular link is in use (i.e., p is the probability that a link is busy). In addition to a utilization percentage or loading, p is also sometimes referred to as an occupancy. The probability that a link is idle is denoted by $q = 1 - p$.

When any one of n parallel links can be used to complete a connection, the composite blocking probability B is the probability that all links are busy†:

$$B = p^n$$

Probability that all links (n) are available $= q^n$

Blocking Probabilities : Lee Graphs

When a series of n links are all needed to complete a connection, the blocking probability is most easily determined as 1 minus the probability that they are all available:

$$B = 1 - q^n$$

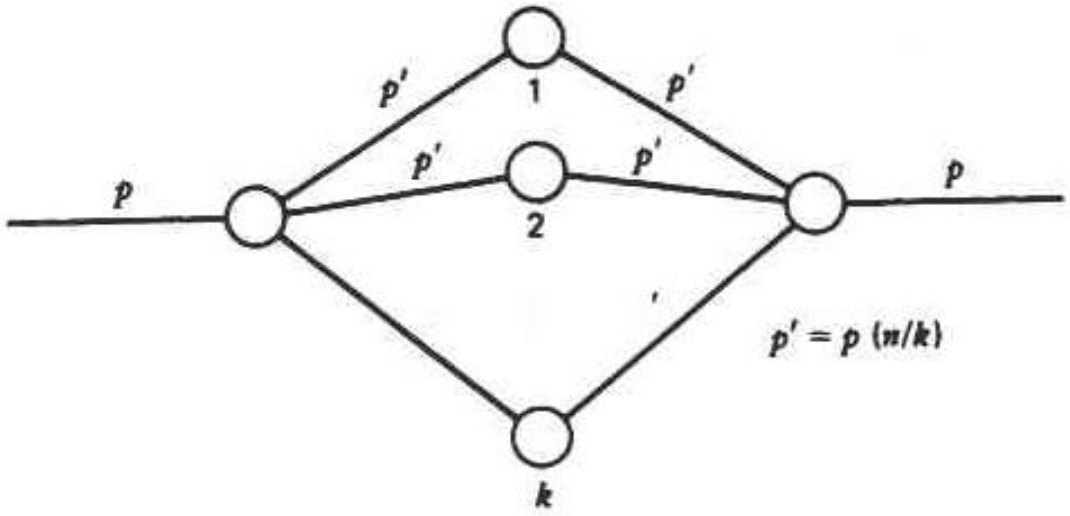


Fig: Probability graph of three stage network.

Blocking Probabilities : Lee Graphs

From the Lee Graph :

- Any particular connection can be established with k different paths : one through each center stage array

Where,

p' = Probability that any interstage link is busy

B = Probability that all paths are busy

= (Probability that an arbitrary path is busy)^k

= (Probability that at least one link in a path is busy)^k

= $\left(1 - q'^2\right)^k$

k = number of center stage array

q' = Probability that any interstage link is idle

Blocking Probabilities : Lee Graphs

If the probability p that an inlet is busy is known, the probability p' that an inter-stage link is busy can be determined as

$$p' = \frac{p}{\beta}$$

Where,

$\beta = \frac{k}{n}$ is the space expansion factor.

Calculate the value p' if there are four center stage arrays for with one inlet. Assume that, the probability of a link is busy is half.

Blocking Probabilities : Lee Graphs

Blocking Probability of three stage switch in terms of inlet utilization p,

$$B = \left(1 - q'^2\right)^k$$

$$B = \left[1 - \left(1 - \frac{p}{B}\right)^2\right]^k$$

TABLE 5.2 Three-Stage Switch Designs for Blocking Probabilities of 0.002 and Inlet Utilization of 0.1

Switch Size, N	n	k	β	Number of Crosspoints	Number of Crosspoints in Nonblocking Design	
128	8	5	0.625	2,560	7,680	($k = 15$)
512	16	7	0.438	14,336	63,488	($k = 31$)
2,048	32	10	0.313	81,920	516,096	($k = 63$)
8,192	64	15	0.234	491,520	4.2 million	($k = 127$)
32,768	128	24	0.188	3.1 million	33 million	($k = 255$)
131,072	256	41	0.160	21.5 million	268 million	($k = 511$)

Blocking Probabilities : Lee Graphs

TABLE 5.3 Three-Stage Switch Designs for Blocking Probabilities of 0.002 and Inlet Utilizations of 0.7

Switch Size N	n	k	β	Number of Crosspoints	Number of Crosspoints in Nonblocking Design	
128	8	14	1.75	7,168	7,680	($k = 15$)
512	16	22	1.38	45,056	63,488	($k = 31$)
2,048	32	37	1.16	303,104	516,096	($k = 63$)
8,192	64	64	1.0	2.1 million	4.2 million	($k = 127$)
32,768	128	116	0.91	15.2 million	33 million	($k = 255$)
131,072	256	215	0.84	113 million	268 million	($k = 511$)

Pathfinding

Determining a path through a single-stage switch is virtually automatic since the necessary crosspoint is uniquely specified by the inlet–outlet pair to be connected. In contrast, availability of more than one path in a multiple-stage switch complicates the path selection process. The call processor of the switch must keep track of which potential paths for a particular connection are available in a state store. A pathfinding routine processes the state store information to select an available path. Whenever a new connection is established or an old one released, the state store is updated with the appropriate information.

Pathfinding Times

Pathfinding operations require the use of common equipment and must therefore be analyzed to determine the rate at which connect requests can be processed. The time required to find an available path is directly dependent on how many potential paths are tested before an idle one is found. Some systems can test a number of paths in parallel and thereby shorten the processing time. Since the expected number of potential paths that must be tested to find an idle path is a function of link utilization, pathfinding times unfortunately increase when the common control equipment is busiest.

Expected number of paths,

$$N_p = \frac{1 - p^k}{1 - p}$$

Example 5.1. What is the expected number of potential paths that must be tested to find an idle path in the three-stage, 8192-line switch defined in Table 5.2?

Time Division Switching

- Time Division Switching (TDS) is a technique used in telecommunications to manage and route data in time-division multiplexed (TDM) networks.
- TDS allows multiple signals to share the same communication channel by assigning each signal a specific time slot in a repeating cycle.

Key Concepts of Time Division Switching:

- Time Slots
- Efficiency in Digital Networks
- Switch Types
- Applications
- Advantages

Time Division Switching

- **Time Slots:** Each user or data stream is assigned a unique time slot within a single communication channel.
- Signals are interleaved in time, allowing multiple connections to coexist without interfering with each other.

- **Efficiency in Digital Networks:** TDS is commonly used in digital telephony, especially in systems like ISDN and cellular networks, where it efficiently utilizes available bandwidth.

Time Division Switching

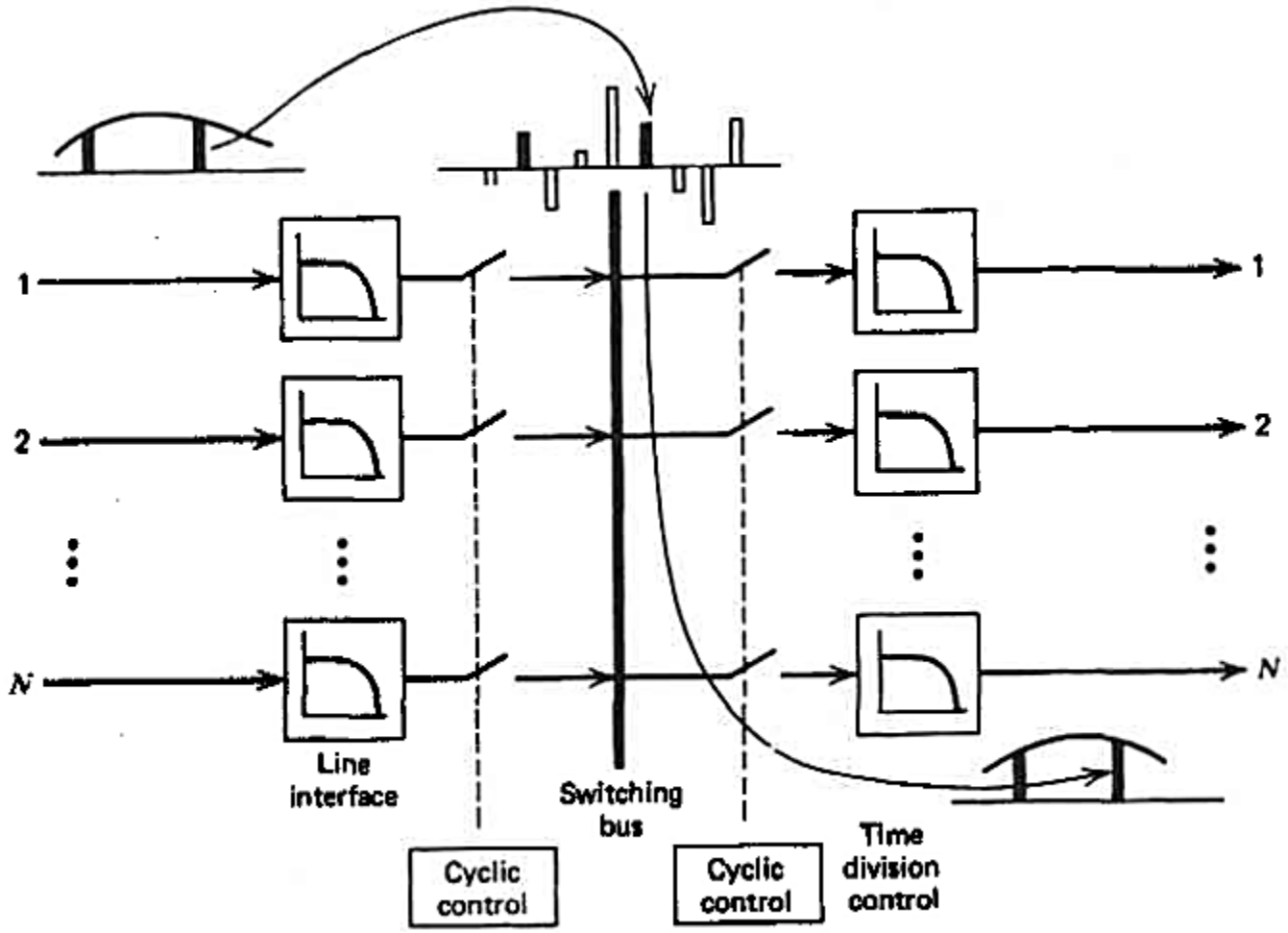
Switch Types:

- **Time-Slot Interchange (TSI)**
- **Time-Multiplexed Space Switch (TMSS**

Applications: TDS is widely applied in digital network switching, particularly in older systems like the Public Switched Telephone Network (PSTN), as well as in more modern digital cellular networks.

Advantages: TDS is efficient for voice communication and is cost-effective in systems where bandwidth is limited and real-time data transmission is necessary.

Time Division Switching



Two Dimensional Switching

Larger digital switches require switching operations in both a space dimension and a time dimension. There are a large variety of network configurations that can be used to accomplish these requirements. To begin with, consider the simple switching structure shown in Figure 5.19. This switch consists of only two stages: a time stage T followed by a space stage S. Thus this structure is referred to a time–space (TS) switch.

The basic function of the time stage is to delay information in arriving time slots until the desired output time slot occurs. At that time the delayed information is transferred through the space stage to the appropriate output link. In the example shown the information in incoming time slot 3 of link 1 is delayed until outgoing time slot 17 occurs. The return path requires that information arriving in time slot 17 of link *N* be delayed for time slot 3 of the next outgoing frame. Notice that a time stage may have to provide delays ranging from one time slot to a full frame.

Two Dimensional Switching

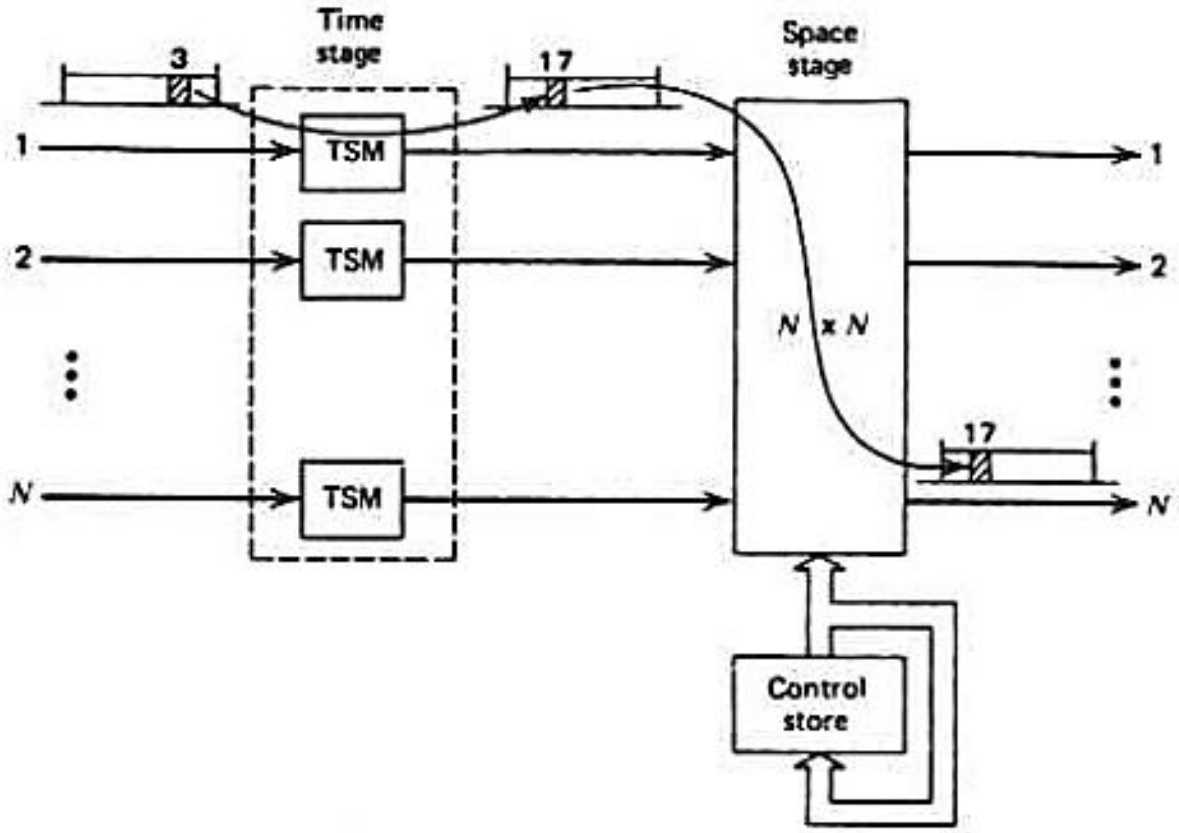


Figure 5.19 Time–space (TS) switching matrix.

Multiple stage Time and Space Switch

As discussed in the preceding section, an effective means of reducing the cost of a time division switch is to multiplex as many channels together as practical and perform as much switching in the time stages as possible. Time stage switching is generally less expensive than space stage switching—primarily because digital memory is much cheaper than digital crosspoints (AND gates). To repeat, the crosspoints themselves are not so expensive, it is the cost of accessing and selecting them from external pins that makes their use relatively costly.

Multiple stage Time and Space Switch

Naturally, there are practical limits as to how many channels can be multiplexed into a common TDM link for time stage switching. When these practical limits are reached, further reductions in the implementation complexity can be achieved only by using multiple stages. Obviously, some cost savings result when a single space stage matrix of a TS or ST switch can be replaced by multiple stages.

A generally more effective approach involves separating the space stages by a time stage, or, conversely, separating two time stages by a space stage. The next two sections describe these two basic structures. The first structure, consisting of a time stage between two space stages, is referred to as a space–time–space (STS) switch. The second structure is referred to as a time–space–time (TST) switch.

STS Switching

A functional block diagram of an STS switch is shown in Figure 5.20. Each space switch is assumed to be a single-stage (nonblocking) switch. For very large switches, it may be desirable to implement the space switches with multiple stages. Establishing a path through an STS switch requires finding a time switch array with an available write access during the incoming time slot and an available read access during the desired outgoing time slot. When each individual stage (S, T, S) is nonblocking, the operation is functionally equivalent to the operation of a three-stage space switch. Hence a probability graph in Figure 5.21 of an STS switch is identical to the probability graph of Figure 5.8 for three-stage space switches. Correspondingly, the blocking probability of an STS switch is

$$B = (1 - q'^2)^k$$

where $q' = 1 - p' = 1 - p/\beta \qquad (\beta = k/N)$

k = number of center-stage time switch arrays

STS Switching

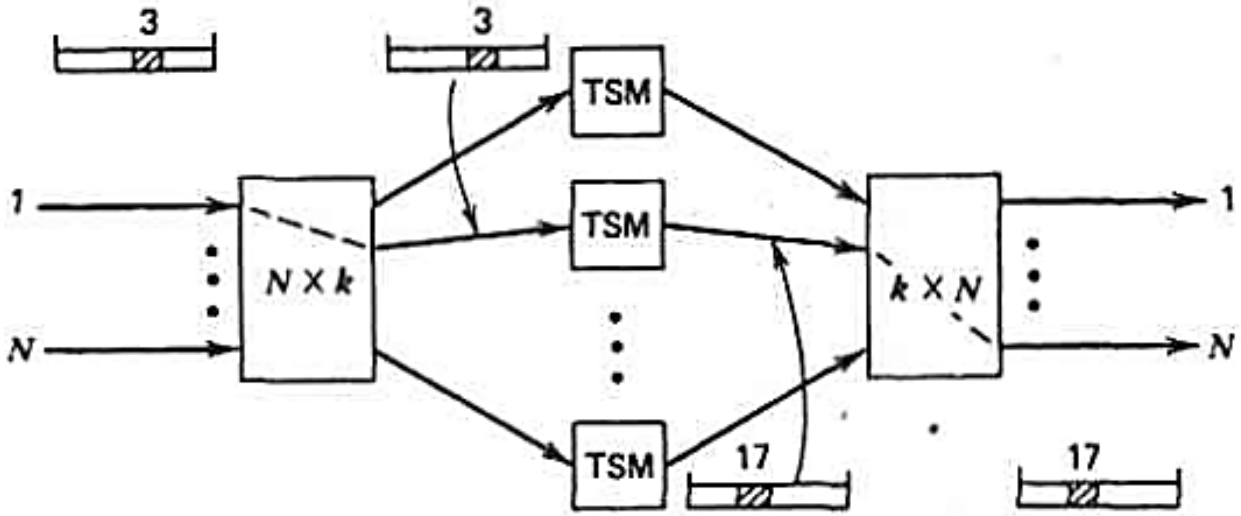


Figure 5.20 Space–time–space (STS) switching structure.

TST Switching

The second form of multiple-stage time–space switch is shown in Figure 5.22— the TST switch. Information arriving in a TDM channel of an incoming link is delayed in the inlet time stage until an appropriate path through the space stage is available. At that time the information is transferred through the space stage to the appropriate outlet time stage where it is held until the desired outgoing time slot occurs. Assuming the time stages provide full availability (i.e., all incoming channels can be connected to all outgoing channels), any space stage time slot can be used to establish a connection. In a functional sense the space stage is replicated once for every internal time slot. This concept is reinforced by the TST probability graph of Figure 5.23.

TST Switching

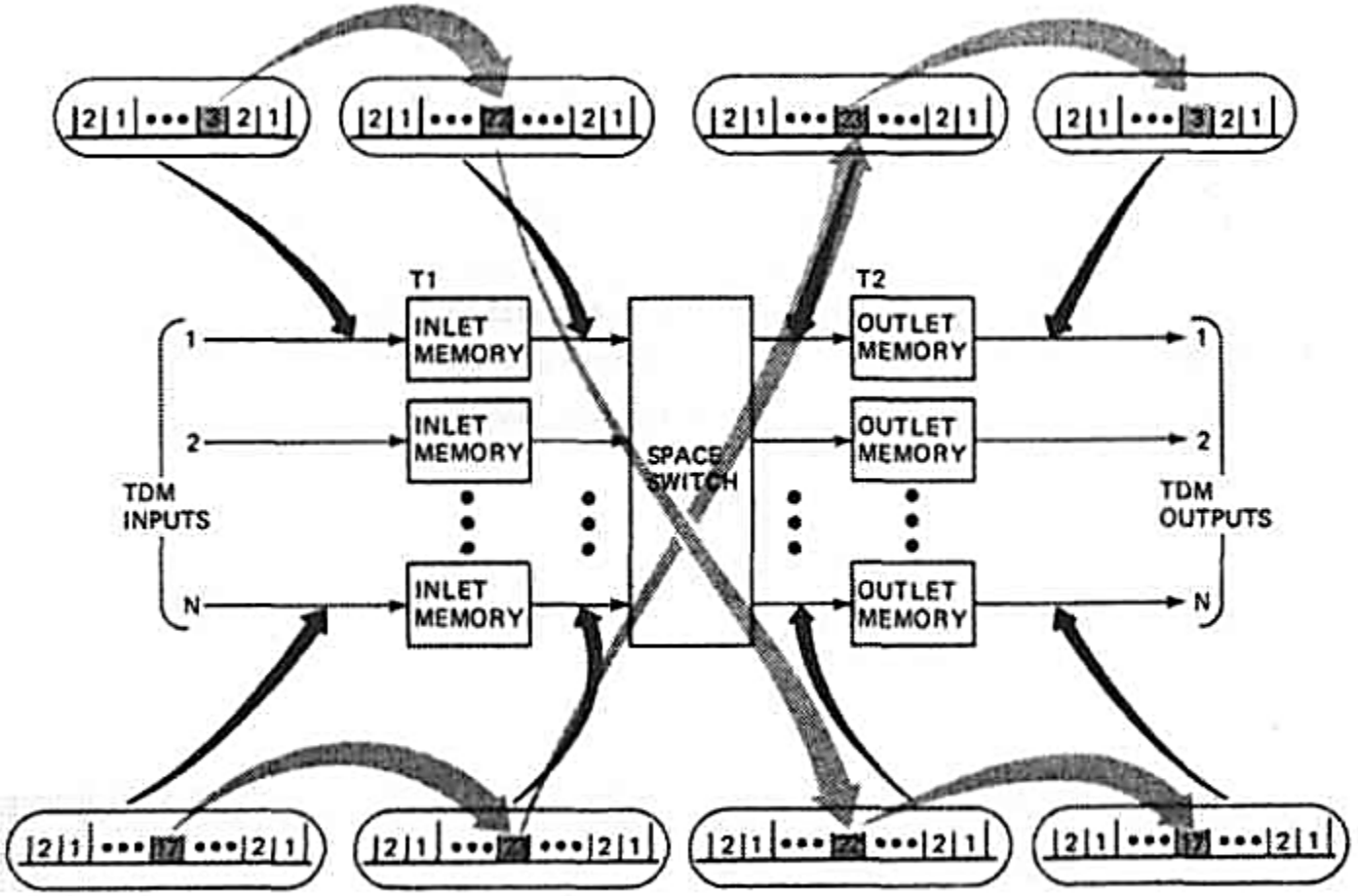


Figure 5.22 Time–space–time (TST) switching structure.

TSSST Switching

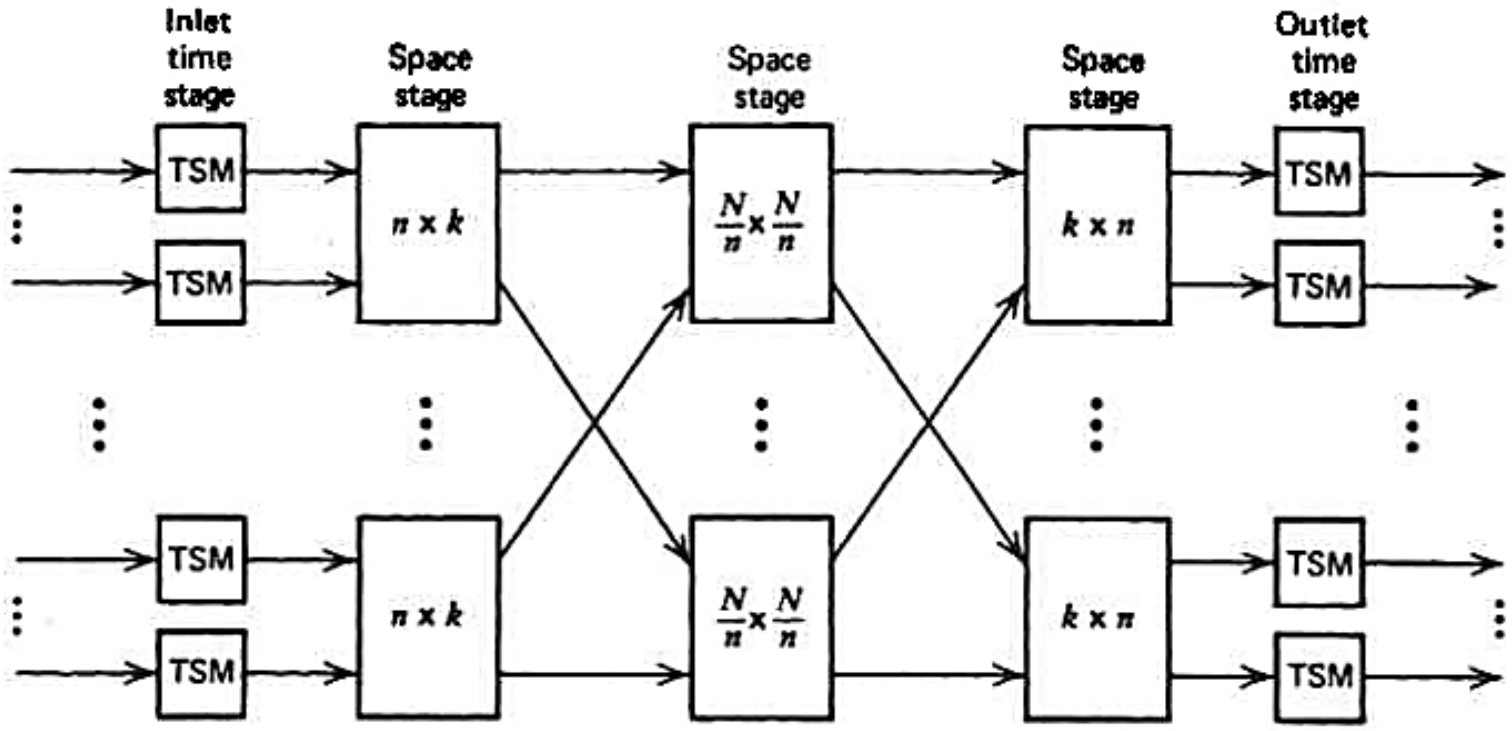


Figure 5.25 Time–space–space–space–time (TSSST) switching structure.